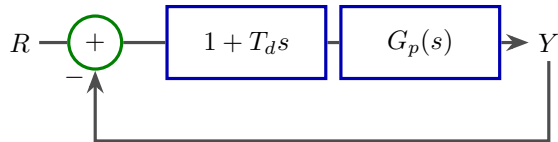


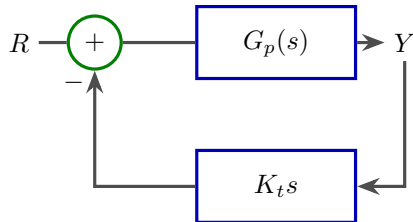
series PD



ζ up

zero added

velocity feedback



$$\zeta_t = \zeta + \frac{1}{2} K_t \omega_n$$

no extra zero